

Behavioural Effects of Supervisory Multi-Agent Control on Operator Workload: A Pilot Study with a Waste-Sorting Robot Simulator

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Abstract

Supervisory control of multi-agent robotic systems (MAS) is expected to reduce operator workload relative to direct manual control, but the size of this effect in realistic task simulations is not well quantified for non-expert users. This pilot study reports a within-subject behavioural comparison of manual versus swarm (supervisory) control in a simulated delta-robot waste-sorting cell. Thirty-four participants from three age/skill groups (school students in grades 7 and 9–11, and Master’s students) each completed both conditions, yielding 93 logged trials. In a paired analysis (Wilcoxon signed-rank, 26 participants who performed both conditions), supervisory control reduced median task completion time from 48.2 s to 19.2 s ($p < 0.001$, matched-pairs rank-biserial $r = 0.95$) and reduced the median number of operator interventions per trial from 43 to 13 ($p < 0.001$, $r = 0.99$). The reduction was consistent across all three groups. Operator skill influenced task time under supervisory control (Master’s students fastest; Kruskal-Wallis $p = 0.009$) but did not significantly affect intervention counts ($p = 0.39$). This is a proof-of-concept study with a non-expert, partly school-age sample in a simulated environment; within these limits, the behavioural benefit of supervisory control is large and consistent.

Keywords: *multi-agent systems; supervisory control; operator workload; human-robot interaction; fan-out; pilot study*

1. Introduction

Industrial automation increasingly relies on multi-agent robotic systems (MAS) supervised by a single human operator who resolves exceptions beyond the agents’ local decision capacity [1]. As fleet size grows, direct teleoperation does not scale [2]: the operator is overwhelmed by asynchronous exception streams. Supervisory control — delegating routine actions to autonomous agents and reserving human attention for exceptions — is the standard mitigation, but its quantitative benefit for non-expert operators in realistic simulations is not well characterised.

This paper reports a pilot study that quantifies the behavioural effect of supervisory (“swarm”) control versus manual control on two workload proxies — task completion time and number of operator interventions — in a simulated waste-sorting MAS, across three age/skill groups.

Contributions.

- A within-subject behavioural pilot dataset (34 participants, 93 trials, three age/skill groups) comparing manual and supervisory control in a MAS waste-sorting simulator.
- Preliminary evidence that supervisory control produces a large, statistically significant reduction in both task time and intervention count for non-expert users.
- A characterisation of how operator skill relates to performance under supervisory control.

2. Related Work

Human supervisory control of autonomous systems involves a tension between agent autonomy and human oversight [1], and direct teleoperation is known not to scale with fleet size [2]. Fan-out — the number of robots one operator can effectively supervise — was introduced as a scalability metric for human-robot teams [2], building on broader principles of efficient human-robot interaction [3]. Frameworks for assessing situation awareness and task performance in human-swarm interaction have also been proposed [4]. The present study contributes behavioural data on the manual-versus-supervisory contrast for non-expert operators, a population less commonly studied than trained operators.

3. System and Task

The testbed is a simulated MAS waste-sorting cell: a conveyor belt served by N delta-robot agents, each with a local machine-vision module and an inverse-kinematics controller. Each agent operates within its own work zone and acts autonomously when its detection confidence exceeds a threshold. When confidence is low (occlusion, deformation, unusual viewing angle) or a physical conflict is detected (e.g., anomalous actuator-current spikes at grasp), the agent places the object on a reject lane and raises an asynchronous interrupt to the central console. The operator's task is to resolve this exception queue.

Two control conditions were implemented and compared:

- **Manual:** the operator directly controls sorting actions.
- **Swarm (supervisory):** agents act autonomously; the operator intervenes only on raised exceptions.

4. Experimental Setup

Participants and design. Thirty-four participants from three groups took part in a simulated waste-sorting task: school students in grade 7 (~13–14 y), school students in grades 9–11 (~15–17 y), and Master's students. The design was within-subject: each participant completed three trials — one manual, one swarm, and a third whose control mode was assigned at random by a coin flip. Only successful trials were logged, giving 93 trials in total; the number of logged trials per participant therefore varies slightly. Manual trials were generally presented before swarm trials, so presentation order was not counterbalanced (see Limitations). Because participant identifiers were not stored, participant boundaries were reconstructed from the session logs by treating each reset of the per-participant trial counter as the start of a new participant, yielding 34 participants (10 / 20 / 4 across the three groups), of whom 26 contributed trials in both control conditions.

Ethics. The study was conducted on the premises of the participating school with the permission of the school administration, during a scheduled physical-education slot, and participation was voluntary. Notification of parents was handled by the school administration; no objections were raised. The activity was treated as a low-risk educational exercise. No personally identifying information (names or similar) was collected at any point — the logs contain only group label, timestamp, and task metrics — so the dataset is anonymous by construction. Explicit written informed consent was not collected, and no institutional ethics-committee approval was obtained; the study is therefore reported as an informal, low-risk classroom pilot rather than a formally approved protocol.

Outcome measures. (i) Task completion time (Duration, s); (ii) number of operator interventions (Clicks). Both are workload proxies; lower is better.

Statistics. Because the design is within-subject, the primary manual-vs-swarm contrast uses the Wilcoxon signed-rank test on per-participant means (participants contributing trials in both conditions), with the matched-pairs rank-biserial correlation r as effect size. Between-group (skill) comparisons use the Kruskal–Wallis test. Distributions are skewed, motivating non-parametric methods throughout.

5. Results

5.1 Automation effect (manual vs. supervisory control)

Supervisory control substantially reduced both workload proxies. In the paired analysis (26 participants with trials in both conditions), median task time fell from 48.2 s (manual) to 19.2 s (swarm) (Wilcoxon signed-rank, $p < 0.001$, $r = 0.95$), and median intervention count fell from 43 to 13 ($p < 0.001$, $r = 0.99$). Both effects are very large, and the direction of change was consistent across groups (Table 1).

Table 1. Per-participant task time and interventions by group and control mode (median [IQR] across participants).

Group	Mode	n (participants)	Task time, s (median [IQR])	Interventions (median [IQR])
Grade 7 (~13–14 y)	Manual	8	61.3 [46–89]	37 [31–67]
Grade 7 (~13–14 y)	Swarm	9	28.0 [20–38]	9 [7–16]
Grades 9–11 (~15–17 y)	Manual	17	45.4 [30–65]	43 [39–71]
Grades 9–11 (~15–17 y)	Swarm	18	20.0 [14–24]	15 [9–22]
Master's students	Manual	4	33.8 [23–44]	44 [24–62]
Master's students	Swarm	4	12.3 [10–13]	15 [10–21]

Group sizes differ between modes because not every participant contributed a logged trial in each mode after the random third trial and the success-only logging.

5.2 Effect of operator skill

Under supervisory control, task time differed significantly across groups (Kruskal–Wallis $H = 9.53$, $p = 0.009$), with Master's students fastest (median 12.3 s) and grade-7 students slowest (median 28.0 s) — a plausible expertise effect on execution speed. In contrast, intervention counts did **not** differ significantly across groups under supervisory control ($H = 1.88$, $p = 0.39$). The reduction in interventions delivered by supervisory control therefore appears broadly accessible to non-expert operators, independent of skill level.

6. Discussion

Within the limits of a small, non-expert, simulated-environment pilot, the behavioural benefit of supervisory control is clear and consistent: delegating routine sorting to autonomous agents reduced median task time by roughly 60% and interventions by roughly 70%, with very large paired effect sizes. This supports the basic premise that the principal workload reduction comes from reducing the rate of required operator actions.

The one skill-dependent effect — faster supervisory execution by Master’s students — is consistent with expertise improving exception-handling speed, but the small Master’s subsample (4 participants) makes this tentative. The absence of a skill-related difference in intervention counts suggests that the reduction in unnecessary actions is a general property of the supervisory paradigm rather than a skill-mediated one, at least for the measures recorded here.

7. Limitations

1. **Sample.** Participants were school students and Master’s students, not professional MAS operators. Two groups are minors. Generalisation to industrial operators is limited.
2. **Sample size and imbalance.** Group sizes are small and uneven (notably only 4 Master’s participants), reducing power for between-group comparisons.
3. **Presentation order.** Manual trials generally preceded swarm trials, so order was not counterbalanced; practice or fatigue effects cannot be fully separated from the control-mode effect.
4. **Simulation.** A simulated conveyor lacks real hardware dynamics, network latency, and real operational stakes.
5. **Workload proxies.** Task time and intervention count are indirect proxies for workload; no direct measure of operator cognitive load was recorded.

8. Conclusion and Future Work

This pilot study provides preliminary evidence that supervisory (“swarm”) control substantially reduces operator workload relative to manual control in a simulated waste-sorting MAS, measured by task time (median 48.2 → 19.2 s) and intervention count (median 43 → 13), with very large paired effect sizes consistent across all three age/skill groups. Operator skill affected supervisory task speed but not intervention counts.

Future work. A natural next step is to complement these behavioural proxies with a direct measure of operator cognitive load across the different control types, and to replicate the study with professional operators on physical hardware, using a counterbalanced design.

References

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